MZUZU UNIVERSITY

FACULTY OF EDUCATION

DEPARTMENT OF MATHEMATICS

SYLLABUS

1	**Programme**	Bachelor of Science (Education)
2	**Subject**	Mathematics
3	**Course Title**	Precalculus
4	**Course Code**	EMT 1101
5	**Level of Study**	One
6	**Prerequisites**	MSCE
7	**Duration**	14 weeks
8	**Lecture Hours per week**	4
9	**Tutorials/Seminars per week**	1
10	**Practical Hours per week**	Nil
11	**Assessment**	Continuous 40% Examination 60%

12 **Aim(s) of the Course**

1 Normalize the knowledge that students arriving possess in Mathematics

2 Develop an intuition in and facility in using mathematical methods

3 Prepare students for subsequent courses in calculus

13 **Objectives of Study**

By the end of the course students should be able to:

1 Use the language of sets to conceptualize mathematical problems

2 Describe the various classes of real numbers

3 Solve equalities and inequalities

4 Graph equations and functions

5 Classify and manipulate the various classes of simple functions

6 Solve optimization problems without the use of calculus

7 Apply the methods learnt in the context of logarithmic, exponential and trigonometric functions

8 Solve systems of linear equations using matrix methods

14 **Topics of study**

1 **Review of sets - Venn diagrams, symbolism**

Venn diagrams: intersection, union, complement
Venn diagrams of word problems
Symbolism of logic argument: statement, proposition, theorem, lemma, implies, if and only if (iff), negation

2 **Algebra and geometry of the real numbers**

Review of exponents and nth roots
Decimal representation, approximation
Sets of real numbers
Absolute value
Polynomials and factoring
Quadratic equations
Inequalities - linear and nonlinear, using absolute values

3 **The Coordinate Plane**

The distance formula and equations of circles
Graphs of equations
Equations of Lines - point-slope formulas
Symmetry in graphs - about the axes, through the origin, in y=x

4 **Functions**

Definition - variables: independent, dependent; domain and range. Venn diagrams, functions by tables
Graph of a function - vertical line test, turning points, maxima, minima, average rate of change
Translations and reflections of graphs of functions
New functions from old - arithmetical operations, composition
Inverse functions - one-to-one, onto, relating the graphs

5 **Polynomial and Rational Functions: applications to optimization**

Definition - variables: independent, dependent; domain and range. Venn diagrams, functions by tables
Graph of a function - vertical line test, turning points, maxima, minima, average rate of change
Linear and quadratic functions
Functions from word problems
Power, polynomial and rational functions – asymptotes

6 **Exponential and Logarithmic functions**

Exponential functions
The number e and function y=e^x
Logarithmic functions
Properties of logarithms
Applications: compound interest, growth and decay

7 **Trigonometric functions**

Circular measure - degrees, turns and radians
The six trigonometric functions of angles
Evaluating trigonometric functions
Trigonometric equations and identities

Trigonometric functions on the real numbers, graphs of trig functions, simple harmonic motion
Analytical trigonometry - right-triangles, trig ratios, sine and cosine rules, addition etc. formulas, inverse trig functions

8 **Systems of equations**

Two linear equations in two unknowns
Matrix formulation and Gaussian elimination
Algebra of matrices and inverse of square matrix
Determinants and Cramer's rule
Nonlinear systems in two variables

15 **Prescribed Texts**

1 Cohen, D. (1994), ***Precalculus with Unit-Circle Trigonometry***, West Publishing

16 **Recommended Texts**

1 Swokowski, E.W. Cole J.A. (1998) ***Precalculus: Functions and Graphs,*** Brooks/Cole Publishing